

Nucleation density programmes the viscoelastic gel point of a percolation network

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Passive microrheology locates the viscoelastic gel point through the Winter–Chambon condition $\alpha = 0.5$. Using three-dimensional random walks in the void of site-percolation lattices, we identify this gel point with the void-percolation transition, $p'_c \simeq 1 - p_c$, and show that a single nucleation-density parameter κ tunes it continuously from 0.68 towards unity. Finite-size scaling yields an effective exponent $\nu \approx 1.6$ with no systematic dependence on κ for $\kappa \leq 0.8$, distinct from static percolation, indicating that nucleation density shifts the gel point while preserving its critical character.

Gelation—the transition of a polymer solution or colloidal suspension from a viscous fluid to a viscoelastic solid—is characterised by a gel point p'_c at which the loss tangent $\tan \delta = G''/G'$ becomes frequency independent and equal to unity [1, 2]. In the Winter–Chambon picture this is the phase angle $\delta = 45^\circ$ and a power-law mean squared displacement (MSD) $\langle r^2(t) \rangle \propto t^{0.5}$, both accessible by passive microrheology through the Generalised Stokes–Einstein Relation (GSER) [3–5]. The gel point is routinely treated as a fixed property of composition—set by concentration, cross-link density, and temperature [6, 7]. What is rarely asked is whether it also depends on *how* the network forms: on the spatial growth dynamics of the cluster itself, which in real colloidal and biopolymer gels range from open fractal aggregates to compact space-filling domains [8–10].

Microrheology has previously been used to locate the gel point of real gels [11, 12], and growth or nucleation rules are known to shift the percolation threshold and even change the order of the transition [13–16]. What has been missing is a single picture in which the passive-microrheology gel point is *identified* with the void-percolation transition and then moved continuously by one morphological parameter—which is what the nucleation density κ provides here.

Here we show, by direct simulation, that the gel point is set by growth dynamics and can be programmed continuously through a single nucleation-density parameter, while its effective critical character is preserved. The approach is grounded in how microrheology actually works: probe particles diffuse through the *void* (fluid) phase of the network, not through the network [17–19]. We therefore place random walkers on the unoccupied sublattice of an $L \times L \times L$ simple cubic lattice, sweep the occupation probability p , and extract the anomalous-diffusion exponent $\alpha(p)$ from $\langle r^2(t) \rangle \propto t^\alpha$; we identify the gel point operationally as the value of p at which $\alpha = 0.5$, equivalently $\delta = 45^\circ$ for a power-law MSD through the GSER [1, 2, 20]. Walkers move to a uniformly chosen nearest neighbour if it is unoccupied and otherwise dwell for a short binding time; the ensemble MSD is accumulated online and α is fitted over the intermediate decade $[L_W/100, L_W/10]$ (see Supplemental Material [21] for all

methods).

Network growth is controlled by a nucleation-density parameter $\kappa \in [0, 1]$. At each increment in p , a fraction $(1 - \kappa)$ of the newly occupied sites are placed as independent random nuclei and the remainder are grown by nearest-neighbour adjacency from the existing cluster frontier. Thus $\kappa = 0$ is classical Bernoulli site percolation (every site an independent seed: maximal nucleation density), $\kappa = 1$ is single-seed compact (Eden [22]) growth, and intermediate κ interpolates between a spatially uniform obstacle field and a compact, clustered network. The seed fraction $1 - \kappa$ is the nucleation density—*independent nuclei per newly occupied site*—so κ indexes that density inversely. We treat κ as a coarse-grained proxy for nucleation conditions that are varied experimentally through cross-linker concentration, thermal-ramp rate, or ionic strength, rather than a parameter set directly in the laboratory.

The gel point is the void-percolation transition.—For random growth ($\kappa = 0$), $\alpha(p)$ falls from Fickian diffusion ($\alpha \approx 1$) through the gel criterion and arrests sharply [Fig. 1]. The crossing is essentially size independent above $L = 200$, extrapolating to

$$p'_c(\infty) \approx 0.681, \quad (1)$$

within one percent of the complementary void-percolation threshold $1 - p_c = 0.6884$ ($p_c = 0.3116$). The identification is geometric: the walkers occupy the void sublattice, itself a site-percolation problem at occupation $1 - p$, which loses spanning connectivity at $p = 1 - p_c$. A small residual offset is expected, because p'_c is defined dynamically through α rather than a static spanning condition; at the incipient void cluster the MSD is $\langle r^2(t) \rangle \propto t^{0.5}$ —the Winter–Chambon criterion. The rheological gel point of a randomly grown network is therefore not an independent parameter but is fixed by the geometry of its void phase; moving it requires changing the network morphology.

Nucleation density tunes the gel point.—Increasing κ (lowering the nucleation density) shifts the $\alpha = 0.5$ crossing monotonically to higher p [Fig. 2]. A compact, connected cluster of a given mass occupies a more localised

region than the same number of randomly placed sites, leaving a larger and better-connected void that percolates only at higher occupation. Finite-size scaling gives thermodynamic gel points rising smoothly with κ : $p'_c(\infty) = 0.681, 0.702, 0.722$ for $\kappa = 0, 0.6, 0.8$, resolved cleanly, rising to $p'_c(\infty) \approx 0.79\text{--}0.80$ at $\kappa = 0.97$ —a broadened crossover point carrying larger uncertainty (see below)—and continuing towards the single-seed Eden limit, where a single compact cluster can fill the lattice before fragmenting the void ($p'_c \rightarrow 1$, treated as an asymptotic bound rather than a finite critical point). A fine κ -grid at fixed L resolves the full curve $p'_c(\kappa)$ [21], monotonic with two regimes separated near a $\sim 1\%$ seed fraction ($\kappa \approx 0.99$): a gradual rise where many nuclei keep the morphology quasi-random, and a steep rise towards unity where single-cluster growth dominates. Across the accessible range the gel point spans $p'_c \approx 0.68\text{--}0.91$, approaching unity only asymptotically in the Eden limit.

One universality class.—Does moving p'_c with κ change the character of the transition? Finite-size scaling across $L = 50\text{--}750$ answers no. The gel point converges by $L \approx 200$, so the shift $p'_c(L) = p'_c(\infty) + aL^{-1/\nu}$ fixes $p'_c(\infty)$ robustly but leaves ν underdetermined; the exponent is instead obtained from two estimators with genuine leverage—the scaling of the transition width $w(L) \sim L^{-1/\nu}$ and the data collapse of $\alpha(p, L)$ [Fig. 3]. Both give $\nu_{\text{eff}} \approx 1.6$, with jackknife intervals that overlap across $\kappa \leq 0.8$ and show no systematic trend in κ :

$$\nu_{\text{eff}} \approx 1.6 \quad (\text{no systematic } \kappa\text{-trend, } \kappa \leq 0.8), \quad (2)$$

the strongly clustered $\kappa \geq 0.97$ conditions lie in the Eden crossover, where the transition broadens and the exponent estimators become unreliable, and are excluded from the estimate [21]. This effective exponent differs markedly from the static 3D site-percolation value $\nu = 0.88$. The difference is expected: p'_c is defined *dynamically*, through the probe-diffusion exponent, rather than through a static order parameter, so the width over which α crosses 0.5 need not scale with the static correlation length. The substantive result is the *absence of a systematic κ -trend* in ν_{eff} : within our resolution, nucleation density moves the gel point without changing its effective critical character. The gel point is thus a programmable but non-universal threshold, not a family of distinct critical behaviours.

Structure confirms dynamics.—The identification is not merely dynamical. Computing the void-percolation threshold directly—the loss of a spanning empty cluster on the same lattices—reproduces the microrheological gel point wherever the network is percolation-like: the geometric and dynamical thresholds coincide to within our resolution for $\kappa \lesssim 0.8$ [Fig. 4]. They separate only for $\kappa \gtrsim 0.85$, the dynamical point lagging the structural one as the void fragments into large finite pockets, marking the same Eden crossover seen in ν_{eff} . That a transport exponent and a connectivity threshold agree where the

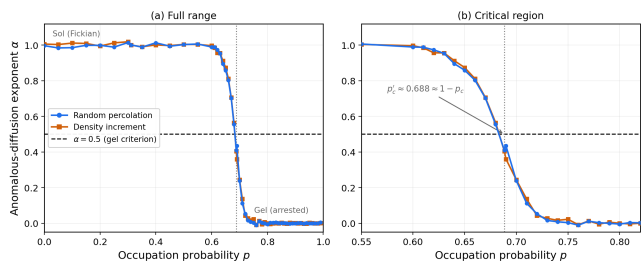


FIG. 1. The gel point is the void-percolation transition. Anomalous-diffusion exponent $\alpha(p)$ for random ($\kappa = 0$) growth: Fickian diffusion ($\alpha \approx 1$) gives way, through the Winter–Chambon criterion $\alpha = 0.5$ (dashed), to arrest. Finite-size scaling places the gel point at $p'_c(\infty) \approx 0.681$, within one percent of the void-percolation threshold $1 - p_c = 0.6884$ ($p_c = 0.3116$).

medium is critical, and part only as it compactifies, is direct evidence that p'_c is the void-percolation transition, dialled by κ .

Finally, the results map directly onto measurable rheology. Because the MSD is a power law over the anomalous window, the GSER yields $G', G'' \propto \omega^\alpha$ with a frequency-independent phase angle $\delta = 90\alpha$; at p'_c this gives $G' \approx G''$ and $\delta \approx 45^\circ$, the Winter–Chambon signature, across the conditions examined [21]. A microrheology experiment on a network whose nucleation density is varied during formation should therefore observe the gel point shift in the direction, and qualitatively by the amount, that $p'_c(\kappa)$ predicts.

These results recast the gel point as a candidate structural design parameter. Rather than a fixed consequence of composition, it is set by the growth dynamics of the network-forming phase and can be dialled across $0.68 \lesssim p'_c \lesssim 0.91$ (with unity as an asymptotic bound) by controlling nucleation density—while, because the effective critical character shows no systematic κ -dependence, the qualitative rheology at the gel point is preserved. The void-percolation baseline $p'_c \simeq 1 - p_c$ supplies the reference; nucleation density supplies the tuning; and finite-size scaling indicates that the tuning is consistent with a single effective critical character. Whether $\nu_{\text{eff}} \approx 1.6$ admits an exact value, and how it relates to the spectral and fractal dimensions of the incipient void cluster, are natural next questions.

[Acknowledgments to be completed.]

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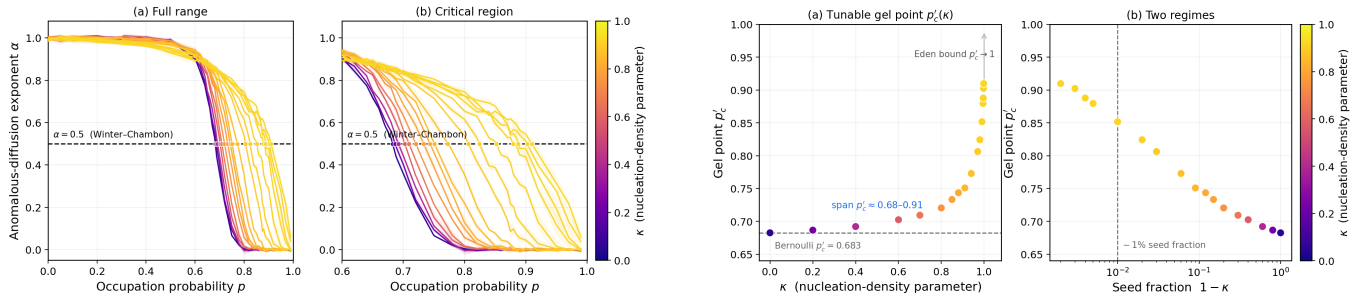


FIG. 2. **Nucleation density tunes the gel point.** (a) $\alpha(p, \kappa)$: the $\alpha = 0.5$ crossing shifts monotonically to higher p as κ increases from 0 (random) towards 1 (single-seed Eden). (b) Gel-point curve $p'_c(\kappa)$, spanning $p'_c \approx 0.68$ – 0.91 over the accessible range, with two regimes separated near a $\sim 1\%$ seed fraction.

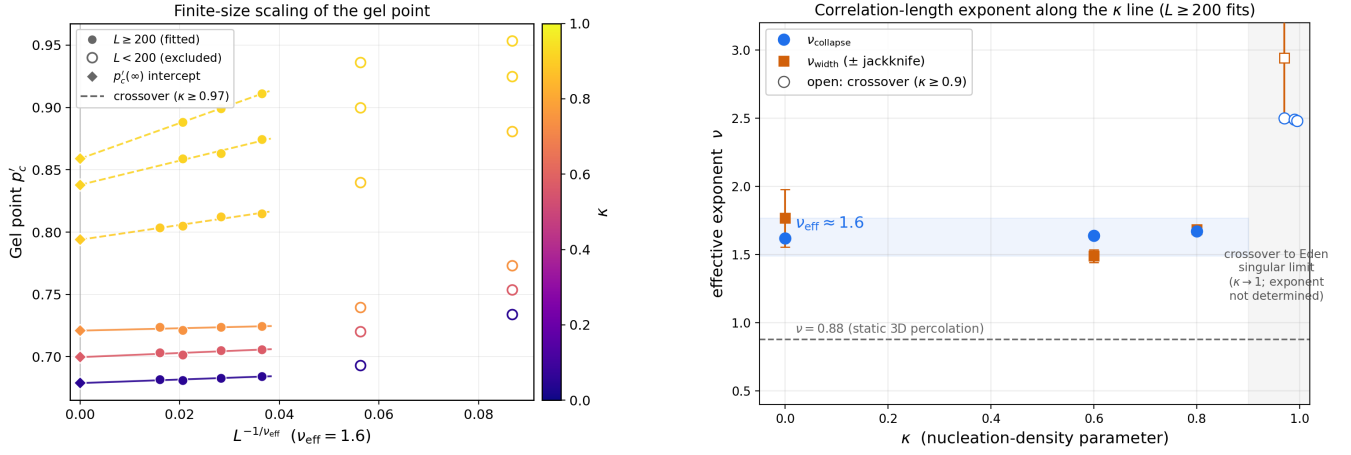


FIG. 3. **A single tunable universality class.** (a) Finite-size scaling $p'_c(L) = p'_c(\infty) + aL^{-1/\nu}$ for each κ (filled: $L \geq 200$ used in fits; open: smaller sizes excluded). (b) Effective exponent $\nu_{\text{eff}}(\kappa)$ from data collapse and transition-width scaling: both agree at $\nu_{\text{eff}} \approx 1.6$, independent of κ and distinct from static 3D percolation ($\nu = 0.88$, dashed). The $\kappa = 0.97$ point (open) lies near the Eden crossover and is excluded from the exponent estimate.

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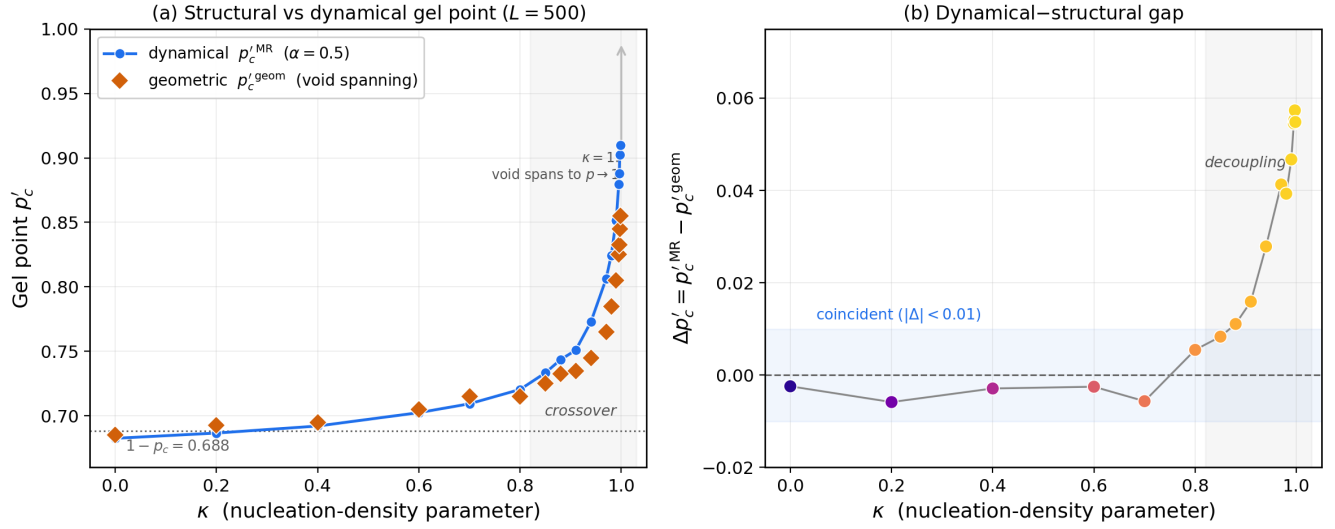


FIG. 4. **The microrheological gel point is the void-percolation threshold.** (a) Dynamical gel point $p_c'^{\text{MR}}$ (the $\alpha = 0.5$ crossing) and the geometric void-spanning threshold $p_c'^{\text{geom}}$ (loss of a face-to-face spanning empty cluster), both at $L = 500$. (b) Their difference $\Delta p'_c = p_c'^{\text{MR}} - p_c'^{\text{geom}}$: the two coincide to within the p -grid resolution for $\kappa \lesssim 0.8$ and decouple through the Eden crossover, the dynamical point lagging the structural one. At $\kappa = 1$ the void of a single compact cluster spans until $p \rightarrow 1$ (no finite- p threshold).